Problem 1a

Segment 2

$$\int_0^1 \frac{\ln(x^4+1)}{x^3}\,\mathrm{d}x$$

Problem 1b

Segment 2

$$\int_{-1}^1 \frac{\ln(x^4+1)}{x^3}\,\mathrm{d}x$$

Problem 2a

Segment 2

$$\int \frac{1}{(x+1)\sqrt{x}+2x}\,\mathrm{d}x$$

Problem 2b

Segment 2

$$\int \frac{1}{(\sqrt{x}+1)x+2\sqrt{x}}\,\mathrm{d}x$$

Problem 3a

Segment 2

$$\int \frac{5x^2+3x-2}{x^3+2x^2}\,\mathrm{d}x$$

Problem 3b

Segment 2

$$\int \frac{(x+1)^2}{(x+2)(x^2+x+1)}\,\mathrm{d}x$$

Problem 4a

Segment 2

$$\int_0^\infty \frac{1}{(x+\sqrt{x^2+1})(1+x^2)}\,\mathrm{d}x$$

Problem 4b

Segment 2

$$\int_0^\infty \frac{1}{(x+\sqrt{x^2+1})(1+x)}\,\mathrm{d}x$$

Problem 5a

Segment 2

$$\int_0^1 \ln\left(1+\sqrt{1+\left(\frac{x^4-1}{2x^2}\right)^2}\right)\,\mathrm{d}x$$

Problem 5b

Segment 2

$$\int_0^\infty \exp\left(-1-\sqrt{1+\left(\frac{x^4-1}{2x^2}\right)^2}\right)\,\mathrm{d}x$$

Problem 6a

Segment 2

$$\int_0^\infty \frac{\cos(x)-\cos^2(x)}{x^2}\ln(x)\,\mathrm{d}x$$

Problem 6b

Segment 2

$$\int_0^\infty \frac{\cos(x)-\cos^2(x)}{x^2}\,\mathrm{d}x$$

Problem 7a

Segment 2

$$\lim_{n\to\infty} \int_1^n \ln\left(1+\frac{e}{n}+\frac{1}{\sqrt[e]{nx}}\right)\,\mathrm{d}x$$

Problem 7b

Segment 2

$$\lim_{n\to\infty} \int_1^n \ln\left(1+\frac{2}{n}+\frac{1}{\sqrt{nx}}\right)\,\mathrm{d}x$$